

Muhammad Nehal

+91-7903999499 | imnehal360@gmail.com | linkedin.com/in/imnehal | github.com/imnehal360

EDUCATION

Indian Institute of Information Technology Manipur

B.Tech in Computer Science and Engineering, CPI: 8.52

Manipur, India

Aug 2023 – Jun 2027

EXPERIENCE

Software Developer Intern – KickBegin

Remote

MERN Stack, Python

May 2026 – Present

- Built a multi-agent **RAG pipeline for AI-powered interviews** using Python microservices and a ChromaDB-backed Embedding Service for semantic question generation.
- Integrated Google **Gemini API** for resume parsing and AI-driven answer evaluation.
- Assisted in deploying the full-stack application on **AWS EC2 using Docker Compose**, helping containerize services, configure the deployment environment, and troubleshoot production issues.

PROJECTS

ResJo AI – AI-Powered Resume & Job Recommendation | [Live](#)

Jun 2026 – Aug 2026

MERN, Gemini API, Tailwind CSS

- Built a Full-Stack AI Career Platform featuring PDF resume parsing, real-time **Gemini-powered conversational resume building**, dynamic **ATS score analysis**, and **personalized job matching**.
- Engineered a Hybrid Job Recommendation Engine integrating Google Gemini **Vector Embeddings (3072-dim) and Cosine Similarity (70% Semantic + 30% Keyword Matching)**, improving candidate-to-job relevance.
- Architected an **automated 72-hour Job Ingestion Pipeline** using RapidAPI (JSearch) and **SHA-256 content hashing** to ensure zero duplicate job listings across multi-role queries in MongoDB Atlas.

MindLens – Multimodal Mental Health Analysis System | [Live](#)

Jan 2026 – Apr 2026

Python, PyTorch, BERT, Librosa, OpenCV, FastAPI, React

- Built a **multimodal AI system leveraging text, audio, and facial data** for detecting mental health conditions.
- Fine Tuned deep learning models (**BERT, CNN, Wav2Vec2**) and applied fusion strategies for unified risk scoring.
- Architected scalable FastAPI endpoints with real-time inference, **SHAP**-based explainability, and **LSTM models** to analyze temporal emotional patterns and detect mental health deviations.

MediSlot – Doctor Appointment Scheduling Platform | [Live](#)

Aug 2025 – Nov 2025

MERN Stack, JWT, Bcrypt, Tailwind CSS

- Designed and built a full-stack, **multi-tenant platform featuring three distinct user workflows (Patient, Doctor, Admin)** to streamline consultation bookings and schedule management.
- Engineered a secure **RESTful API** using Node.js, Express, and MongoDB, **integrating Stripe and Razorpay payment** gateways with webhook-verified server-side endpoints to process, verify, and secure consultation fees.
- Built an algorithm for doctor availability with automatic slot allocation, cancellation, and double-booking prevention.

TECHNICAL SKILLS

Languages: C++, C, Python, SQL, JavaScript

Backend & APIs: REST APIs, Node.js, Express.js, FastAPI

Frontend: React.js, HTML5, CSS, Tailwind CSS, React Router

Machine Learning: ML Algorithms Scikit-learn, Pandas, NumPy, Matplotlib, OpenCV, Librosa

Deep Learning: Neural Networks, CNN, RNN, LSTM, Transformer, PyTorch, TensorFlow

Databases: MongoDB, MySQL

Tools & DevOps: Git, Github, CI/CD Pipeline, Docker, AWS EC2, Cloudinary, Postman, Render

Relevant Coursework: Data Structures and Algorithms, Database Management System, Operating System, Object-Oriented Programming, Computer Networks

ACHIEVEMENTS

- Secured **99th rank** and awarded the **Bronze Medal at the ICPC Asia Taiwan Regional Contest**.
- Secured **732nd rank** out of **80,000+ teams** in the Amazon ML Challenge 2025.
- Achieved a **Top 8% rank** among 750+ students in DSA problem solving on GeeksforGeeks.
- Solved **650+ problems** on LeetCode and Codeforces.